The spectral measure of an elliptic orbital integral, via Arthur's local trace formula

Draft

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Abstract

This AI-aided paper revises and updates the first author's earlier paper [Jo92], using a method different from the one used in the analogous paper [JC26]. Let G be an unramified semisimple group over a p-adic field, K a hyperspecial maximal compact subgroup, and γ a regular elliptic element. Taking one test function spherical and the other supported on the elliptic set, we specialize Arthur's local trace formula. On the geometric side only the orbital integrals of G survive. On the spectral side only two kinds of terms survive: the unramified principal series of G, weighted by a weighted character of the elliptic test function, and Arthur's elliptic tempered virtual characters with K-fixed vectors. We deduce that the spherical orbital integrals of γ are given by a finite measure on the unitary unramified characters. Its absolutely continuous part has density a weighted character. Its singular part is supported on subtori where the R-groups of the unramified principal series are non-trivial, and it vanishes when the center of G is connected. For $SL(n)$ the singular part is given by κ-orbital integrals. We compare the result with the truncation method of the companion paper [JC26], and work out $PGL(2)$, $SL(2)$, $PGL(3)$ and $SL(3)$.

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*Claude is an AI model made by Anthropic. This is a working draft. Statements are labeled as proved here, quoted (with a reference), or expected; every "Status" paragraph says which, and what remains to be checked.

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1 Introduction

The problem. Let F be a p-adic field with ring of integers $\mathcal{O}$ and residue field of order q. Let G be a connected unramified semisimple group over F, and $K \subset G(F)$ a hyperspecial maximal compact subgroup. Let $\widehat{T}^u$ be the compact torus of unitary unramified characters of a minimal Levi subgroup M_0, with Weyl group W. For f in the spherical Hecke algebra $\mathcal{H}(G,K)$ let $f^\vee$ be its Satake transform, a W-invariant function on $\widehat{T}^u$, and let $d\mu$ be the Plancherel measure on $\widehat{T}^u/W$. For $\gamma \in G(F)$ regular elliptic, the orbital integral $f \mapsto O_\gamma(f) = \int_{G(F)} f(g^{-1}\gamma g)\,dg$ is a linear form on $\mathcal{H}(G,K)$, i.e. on W-invariant Laurent polynomials on $\widehat{T}^u$. We ask what kind of object it is:

$$O_\gamma(f) = \int_{\widehat{T}^u/W} f^\vee(\chi)\,dI_\gamma(\chi) \qquad (f \in \mathcal{H}(G,K)), \tag{1}$$

with dI_γ a measure, and if so which measure.

Route A. The companion paper [JC26] (a revision of [Jo92]) attacks (1) directly. It truncates the integral over g to a finite union $G(T)$ of double cosets, and expands the spherical function in matrix coefficients. It then uses Casselman's asymptotics and Brion's theorem to write the truncated kernel $I^T(\chi,\gamma)$ as an exponential polynomial in T. The T-independent part $I(\chi,\gamma)$ is an explicit rational function of χ. Under a regularity hypothesis (R) on the oscillating terms, [JC26, Thm. 4.5] proves $dI_\gamma = I(\chi,\gamma)\,d\mu$. The hypothesis holds for $PGL(2)$. It fails for $SL(2)$,

where dI_γ has an atom at the unramified quadratic character. It also fails for $PGL(3)$, where nevertheless dI_γ is absolutely continuous [JC26, App. B]. So (R) is not the right condition, and [JC26, Conj. 4.6] predicts the general shape of dI_γ.

Route B. Here we derive (1) from Arthur's local trace formula [Ar89a, Ar91], in the form given by his theory of elliptic tempered characters [Ar93]. Arthur's formula is an identity of two expansions of one distribution on $G(F) \times G(F)$. For $f = f_1 \otimes f_2$ it reads, schematically,

$$\sum_M c_M \int_{\Gamma_{\mathrm{ell}}(M)} J_M(\gamma, f)\, d\gamma = \sum_M c_M \int_{\Pi(M)} J_M(\pi, f)\, d\pi,$$

with weighted orbital integrals on the left and weighted characters on the right. Take $f_1 = f$ spherical and $f_2 = \phi$ supported on the regular elliptic set. Then almost everything vanishes.

- On the geometric side only $M = G$ survives, and it is the elliptic orbital integral $I(f, \phi)$ (§4).
- On the spectral side, a weighted character of the spherical function f vanishes unless its weight is trivial (Lemma 2.2). This rests on the fact that the canonically normalized unramified intertwining operators fix the spherical vector. A character of ϕ vanishes on properly induced representations. What remains is a sum over the unitary unramified characters χ and over the finitely many elliptic tempered virtual characters with K-fixed vectors (§5).

Results. The statements below are proved here modulo the normalizations listed in §11, which are being checked.

Theorem A (existence)**.** *For every $\phi \in C_c^\infty(G_{\mathrm{ell}})$ the linear form $f^\vee \mapsto I(f, \phi)$ is given by a finite complex measure dI_ϕ on $\widehat{T}^u/W$.*

Theorem B (the absolutely continuous part)**.** $dI_\phi = I_B(\chi, \phi)\, d\mu(\chi) + dI_\phi^{\mathrm{sing}}$*, where*

$$I_B(\chi, \phi)\, \rho(\chi) = c_G\, J_{M_0}(\chi^{-1}, \phi), \qquad c_G = (-1)^{\dim \mathfrak{a}_0^G} |W|^{-1},$$

ρ is the Plancherel density, and J_{M_0} is Arthur's weighted character for the canonically normalized unramified intertwining operators (§2.3). In particular $I_B\rho$ is a smooth function on $\widehat{T}^u$.

Theorem C (the singular part)**.** *dI_ϕ^{sing} is a finite sum of smooth measures on subtori of $\widehat{T}^u/W$, one for each elliptic tempered virtual character, with K-fixed vectors, of a Levi subgroup $M \supsetneq M_0$ (§5.3). The subtorus attached to M has dimension $\dim \mathfrak{a}_M^G$. If all R-groups of the unitary unramified principal series are trivial, for instance if $Z(G)$ is connected, then $dI_\phi^{\mathrm{sing}} = 0$.*

Theorem D ($SL(n)$)**.** *For $G = SL(n)$, $dI_\gamma^{\mathrm{sing}}$ is the sum over the non-trivial unramified κ of the push-forwards of the stable spectral measures of the elliptic endoscopic groups, as in [JC26, Conj. 4.6(c)].*

The comparison with Route A is Conjecture 7.1: the function I_B of Theorem B equals the T-independent part I of the truncated kernel of [JC26]. If it holds, Route A gives closed formulas for the density, and Route B explains why it is smooth and what replaces it where (R) fails.

Why this is useful. Arthur's local trace formula is one of the deepest tools of local harmonic analysis, and most of its terms are hard to compute. In the case treated here every surviving term can be made explicit. The paper is meant both as an independent proof of (1) and as an introduction to Arthur's local theory through a concrete case.

Plan. §2 collects the unramified background: normalized intertwining operators, R-groups, and weighted characters. §3 states Arthur's local trace formula. §4 and §5 specialize its two sides. §6 assembles Theorems A–C. §7 compares the result with Route A. §8 treats $SL(n)$. §9 works out examples. §10 reviews the literature before and after 1997. §11 lists what remains to be checked, and Appendix A is a glossary.

2 Unramified background

This section collects the facts about unramified principal series that the specialization in §§4–5 needs. The essential point is Lemma 2.2: with the canonical unramified normalization, a weighted character of a spherical function has no weight.

2.1 Notation

We use the notation of [JC26, §2]. A is a maximal split torus, $M_0 = Z_G(A)$, $B = M_0N_0$ a Borel subgroup, W the Weyl group, $\mathfrak{a}_0 = X_*(A) \otimes \mathbb{R}$, and $\mathcal{P}(M)$ the set of parabolic subgroups with Levi component M. For a character χ of $M_0(F)$ and $P \in \mathcal{P}(M_0)$, $I_P(\chi)$ is the normalized induced representation, realized on functions on K (the compact picture). Φ_K is the spherical vector, equal to 1 on K. For unramified χ write $\chi_\alpha = \chi \circ \alpha^\vee$, an unramified character of $F^\times$, and $L(s, \chi_\alpha) = (1 - \chi_\alpha(\pi) q^{-s})^{-1}$. To keep formulas short we assume that G is split. The quasi-split unramified case changes only the local factors attached to roots.

2.2 Normalized intertwining operators

For $P, P' \in \mathcal{P}(M_0)$ let $J_{P'|P}(\chi) : I_P(\chi) \to I_{P'}(\chi)$ be the standard intertwining operator, defined by a convergent integral for χ in a cone and by meromorphic continuation in general. Gindikin–Karpelevich and Casselman [Cas80, Thm. 3.1] give, in the compact picture,

$$J_{P'|P}(\chi)\Phi_K = \prod_{\alpha} c_\alpha(\chi)\, \Phi_K, \qquad c_\alpha(\chi) = \frac{1 - q^{-1}\chi_\alpha(\pi)}{1 - \chi_\alpha(\pi)} = \frac{L(0, \chi_\alpha)}{L(1, \chi_\alpha)}, \tag{2}$$

the product over the reduced roots α that are positive for P and negative for P'. Define

$$r_{P'|P}(\chi) = \prod_{\alpha} \frac{L(0, \chi_\alpha)}{L(1, \chi_\alpha)}, \qquad R_{P'|P}(\chi) = r_{P'|P}(\chi)^{-1} J_{P'|P}(\chi). \tag{3}$$

For unramified data these are Langlands' normalizing factors [La76], with the ε-factors equal to 1 for an additive character of conductor $\mathcal{O}$. They satisfy Arthur's conditions on normalizing factors: multiplicativity, $R_{P|P'} R_{P'|P} = 1$, unitarity and holomorphy on the unitary axis, and compatibility with the K-action. By (2),

$$R_{P'|P}(\chi)\Phi_K = \Phi_K \qquad \text{for all } P, P' \in \mathcal{P}(M_0) \text{ and all } \chi. \tag{4}$$

Status 2.1. For split G this is standard. Arthur's conditions on normalizing factors, and his canonical normalization [Ar98], have not yet been compared in detail with (3); see §11, item (N1).

2.3 Weighted characters

Fix $P \in \mathcal{P}(M_0)$ and unitary χ. For $Q \in \mathcal{P}(M_0)$ and $\Lambda \in i\mathfrak{a}_0^*$ put

$$\mathcal{R}_Q(\Lambda, \chi, P) = R_{Q|P}(\chi)^{-1} R_{Q|P}(\chi_\Lambda),$$

where $\chi_\Lambda = \chi \cdot e^{\Lambda(H_{M_0}(\cdot))}$. Arthur shows that $\{\mathcal{R}_Q(\Lambda, \chi, P)\}_Q$ is a (G, M_0)-family of operators [Ar81, §7]. More generally, for $M \supseteq M_0$ and a Levi $L \supseteq M$ there is an operator $\mathcal{R}_M^L(\chi, P)$, the limit at $\Lambda = 0$ of the associated sum over $\mathcal{P}^L(M)$ [Ar81, §6]. The *weighted character* of $f \in C_c^\infty(G(F))$ is

$$J_M(\chi, f) = \operatorname{tr}\big(\mathcal{R}_M(\chi, P)\, I_P(\chi, f)\big).$$

It does not depend on P. For $M = G$ it is the character $\operatorname{tr} I_P(\chi, f)$. The same construction applies to every tempered representation of a Levi subgroup.

Lemma 2.2. *Let $f \in \mathcal{H}(G, K)$, let χ be unramified, and let $M \subsetneq L$ be Levi subgroups containing M_0. Then $J_M^L(\chi, f) = 0$, where J_M^L is formed with the normalization (3).*

Proof. Since f is bi-K-invariant, $I_P(\chi, f)$ maps $I_P(\chi)$ into the line $\mathbb{C}\Phi_K$ and kills the K-invariant complement of that line. So $I_P(\chi, f) = f^\vee(\chi)\, e_K$, with e_K the projection onto $\mathbb{C}\Phi_K$, and $J_M^L(\chi, f) = f^\vee(\chi)\langle \mathcal{R}_M^L(\chi, P)\Phi_K, \Phi_K^\vee\rangle$. By (4), $\mathcal{R}_Q(\Lambda, \chi, P)\Phi_K = \Phi_K$ for every Q and Λ. So on Φ_K the family is the constant (L, M)-family 1. The limit attached to a constant (L, M)-family is 0 when $M \neq L$, because $\sum_{Q \in \mathcal{P}^L(M)} \theta_Q(\Lambda)^{-1} = 0$ [Ar81, Lemma 6.2]. Hence $\mathcal{R}_M^L(\chi, P)\Phi_K = 0$. □

Remark 2.3. Lemma 2.2 depends on the normalization. With unnormalized operators, the weighted characters of spherical functions would carry the logarithmic derivatives of the c-functions. These are exactly the terms that appear in the T-independent part of the truncated kernel of Route A; compare §7.

2.4 R-groups and the unramified elliptic set

For unitary unramified χ let $W_\chi = \{w \in W : w\chi = \chi\}$, and let W_χ° be the subgroup generated by the reflections s_α with $\chi_\alpha = 1$. The Knapp–Stein R-group is $R_\chi = W_\chi / W_\chi^\circ$. It governs the decomposition of $I_P(\chi)$: the commuting algebra is a twisted group algebra of R_χ, which for unramified principal series is untwisted [Ke82]. For Chevalley groups Keys computed R_χ [Ke82]. In particular:

- if the derived group of the dual group $\hat{G}$ is simply connected (equivalently $Z(G)$ is connected), then $R_\chi = 1$ for all unitary unramified χ;
- for $G = SL(n)$, R_χ is cyclic of order dividing n, and $R_\chi \neq 1$ exactly on a finite union of subtori of $\widehat{T}^u$. For $SL(2)$ this is the point χ_0, the unramified quadratic character.

An element $r \in R_\chi$, represented by $w \in W_\chi$, is *elliptic* if w has no non-zero fixed vector in $\mathfrak{a}_0^G$. Arthur's elliptic tempered virtual characters with K-fixed vectors are exactly the virtual characters $\tau = (\chi, r)$ with r elliptic (§5.3).

3 Arthur's local trace formula

This section states the parts of Arthur's local trace formula that we use. We quote; nothing here is new.

3.1 The truncated kernel

The group $G(F) \times G(F)$ acts on $L^2(G(F))$ by $(R(x_1, x_2)\psi)(y) = \psi(x_1^{-1} y x_2)$. For $f = f_1 \otimes f_2$ the operator $R(f)$ has kernel

$$K_f(x, y) = \int_{G(F)} f_1(u) f_2(x^{-1} u y)\, du.$$

Its restriction to the diagonal is not integrable in general. Arthur multiplies it by the characteristic function $u(x, T)$ of a large compact subset of $G(F)$, defined by a point T in the positive chamber through the convex hull of $W \cdot T$. He shows that $J^T(f) = \int_{G(F)} K_f(x, x) u(x, T)\, dx$ is asymptotic to a polynomial in T [Ar89a, Ar91]. Its constant term $J(f)$ is a distribution on $G(F) \times G(F)$.

3.2 The geometric and spectral expansions

Arthur proves two expansions of $J(f)$ [Ar91, Theorems on pp. ?]:

$$J(f) = \sum_{M \in \mathcal{L}} \frac{|W_0^M|}{|W_0^G|} (-1)^{\dim \mathfrak{a}_M^G} \int_{\Gamma_{\mathrm{ell}}(M)} J_M(\gamma, f)\, d\gamma, \tag{5}$$

$$J(f) = \sum_{M \in \mathcal{L}} \frac{|W_0^M|}{|W_0^G|} (-1)^{\dim \mathfrak{a}_M^G} \int_{\Pi_{\mathrm{temp}}(M)} J_M(\pi, f)\, d\pi. \tag{6}$$

Here $\mathcal{L}$ is the set of Levi subgroups containing M_0, and $\Gamma_{\mathrm{ell}}(M)$ is the set of elliptic regular conjugacy classes of $M(F)$. The weighted orbital integral $J_M(\gamma, f)$ is

$$J_M(\gamma, f) = |D(\gamma)| \int_{M_\gamma \backslash G} \int_{M_\gamma \backslash G} f_1(x_1^{-1} \gamma x_1) f_2(x_2^{-1} \gamma x_2)\, v_M(x_1, x_2)\, dx_1\, dx_2,$$

with v_M the weight of the (G, M)-family $e^{-\Lambda(H_P(x_1) - H_{\bar{P}}(x_2))}$. In (6), $J_M(\pi, f)$ is a weighted character of $G(F) \times G(F)$ attached to $\pi \otimes \pi^\vee$. The measure $d\pi$ and the precise domain of integration are as in [Ar91].

3.3 The elliptic form

In [Ar93] Arthur reorganizes the spectral side using the virtual characters $\tau \in T_{\mathrm{disc}}(M)$: triples (L, σ, r) with σ a square-integrable representation of a Levi $L \subseteq M$ and r in the R-group of σ in M, up to equivalence. The elliptic ones, $T_{\mathrm{ell}}(M)$, are those for which r is elliptic in the sense of §2.4. The spectral side becomes a sum over M of integrals over $T_{\mathrm{disc}}(M)$, with explicit constants. We need its form for $f = f_1 \otimes f_2$ with f_2 cuspidal (§5).

3.4 Splitting

For $f = f_1 \otimes f_2$ the weighted characters split [Ar88], [Ar91, §?]:

$$J_M(\pi, f_1 \otimes f_2) = \sum_{L_1, L_2 \in \mathcal{L}(M)} d_M^G(L_1, L_2)\, J_M^{L_1}(\pi, f_{1,Q_1})\, J_M^{L_2}(\pi^\vee, f_{2,Q_2}). \tag{7}$$

Here $d_M^G(L_1, L_2)$ is non-zero only if $\mathfrak{a}_M^{L_1} \oplus \mathfrak{a}_M^{L_2} = \mathfrak{a}_M^G$, $d_M^G(M, G) = 1$, f_{i,Q_i} are constant terms, and the $Q_i \in \mathcal{P}(L_i)$ are fixed by a choice of a generic point. The same holds for weighted orbital integrals.

Status 3.1. The constants in (5)–(6), the exact domain and measure on the spectral side, and the form of (7) in the local trace formula (as opposed to the invariant trace formula of [Ar88]) are being checked against [Ar91, Ar93]; see §11, item (N2). The theorem numbers in [Ar91] are to be supplied.

4 The geometric side

Let $f_1 = f \in \mathcal{H}(G, K)$, and let $f_2 = \phi \in C_c^\infty(G(F))$ be supported in the regular elliptic set G_{ell}.

Proposition 4.1. *With* $\mathbf{f} = f \otimes \phi$*:*

(a) $x \mapsto K_{\mathbf{f}}(x, x)$ *has compact support, so* $J^T(\mathbf{f})$ *is constant for* T *large, and*

$$J(\mathbf{f}) = \int_{G(F)} \int_{G(F)} f(u)\phi(x^{-1}ux)\, du\, dx = I(f, \phi),$$

the distribution of [JC26, (1.1), (4.8)];

(b) in (5) the terms with $M \neq G$ *vanish, and the term* $M = G$ *is the Weyl integration formula for* $I(f, \phi)$*.*

Proof. (a) For ϕ supported on G_{ell}, the set of x with $\phi(x^{-1}ux) \neq 0$ for some u in the support of f is compact, because centralizers of elliptic elements are compact (Kazhdan [Ka86, §4]; [JC26, Lemma 4.4]). So $K_{\mathbf{f}}(x, x)$ has compact support, and $u(x, T) = 1$ on it for T large.

(b) Let $M \neq G$ and $\gamma \in \Gamma_{\mathrm{ell}}(M)$. If γ is regular in G, its centralizer in G is a maximal torus of M, which contains the non-trivial split torus A_M; so γ is not elliptic in G. Hence $\phi(x^{-1}\gamma x) = 0$ for all x, and $J_M(\gamma, \mathbf{f}) = 0$. The γ that are singular in G form a set of measure zero in $\Gamma_{\mathrm{ell}}(M)$. For $M = G$ the weight is 1, and $J_G(\gamma, \mathbf{f}) = |D(\gamma)| O_\gamma(f) O_\gamma(\phi)$ with the measures of [JC26, (4.10)]. □

5 The spectral side

5.1 Splitting with a spherical and an elliptic function

Proposition 5.1. *Let* $f \in \mathcal{H}(G, K)$ *and* $\phi \in C_c^\infty(G_{\mathrm{ell}})$*, and let* π *be a tempered representation of* $M(F)$ *induced from a unitary unramified character* χ *of* $M_0(F)$*, or more generally any* $\tau \in T_{\mathrm{disc}}(M)$*. Then*

$$J_M(\pi, f \otimes \phi) = f^\vee(\chi)\, J_M(\pi^\vee, \phi).$$

If π *has no* $(K \cap M)$*-fixed vector, both sides vanish.*

Proof. In (7), the factor $J_M^{L_1}(\pi, f_{1,Q_1})$ vanishes for $L_1 \neq M$ by Lemma 2.2, applied to the constant term $f_{1,Q_1} \in \mathcal{H}(L_1, K \cap L_1)$. For $L_1 = M$ the condition $\mathfrak{a}_M^M \oplus \mathfrak{a}_M^{L_2} = \mathfrak{a}_M^G$ forces $L_2 = G$, and $d_M^G(M, G) = 1$. The remaining factor is $J_M^M(\pi, f_{1,Q}) = \operatorname{tr} \pi(f_{1,Q}) = \operatorname{tr} I_Q(\pi, f)$, which equals $f^\vee(\chi)$ if π is a constituent of an unramified principal series with a $K \cap M$-fixed vector, and 0 if it has none. □

5.2 Cuspidality of ϕ

Lemma 5.2. *If ϕ is supported on G_{ell}, then $\operatorname{tr}\sigma(\phi) = 0$ for every representation σ of finite length induced from a proper parabolic subgroup.*

Proof. By van Dijk's formula [vD72], the character of a properly induced representation is supported on the conjugates of the Levi subgroup, which meet G_{ell} in a set of measure zero. □

So ϕ is cuspidal, and in the elliptic form of the spectral side only elliptic data survive in the ϕ-factor.

5.3 Which virtual characters survive

Combining Proposition 5.1 and Lemma 5.2 with Arthur's elliptic form, the spectral side of $J(f \otimes \phi)$ is a sum over $M \in \mathcal{L}$ of integrals, over the virtual characters $\tau \in T_{\mathrm{disc}}(M)$ with $K \cap M$-fixed vectors, of $f^\vee(\chi_\tau) J_M(\tau^\vee, \phi)$, where χ_τ is the unramified character underlying τ. We now list these τ.

Lemma 5.3. *A square-integrable representation of $L(F)$, L a Levi subgroup of G with $\dim \mathfrak{a}_{M_0}^L > 0$, has no vector fixed by the hyperspecial subgroup $K \cap L(F)$.*

Sketch. A $K \cap L$-spherical irreducible representation is a constituent of an unramified principal series $I^L(\chi)$, and its spherical matrix coefficient is the spherical function Γ_χ^L. By Macdonald's formula, $|\Gamma_\chi^L(a)|$ is bounded below by a non-zero multiple of $\delta_B^{1/2}(a)$ on a cone in A^-, for every χ. By the volume formula $\mathrm{vol}(KaK) = \delta_B(a)^{-1} Q_a$, this is not square-integrable. □

Status 5.4. The lower bound in the sketch needs care at non-regular χ, where the terms of Macdonald's formula can cancel. The statement is presumably in the literature (Casselman's criterion for square-integrability [Cas95] gives it directly, since the exponents of a spherical representation lie in $\delta_B^{1/2} \cdot \widehat{T}^u$ up to W). To be completed.

By Lemma 5.3, in a triple $\tau = (L, \sigma, r)$ with a K-fixed vector we must have $L = M_0$ and $\sigma = \chi$ a unitary unramified character. So:

- $M = M_0$: $T_{\mathrm{disc}}(M_0) = \widehat{T}^u$, and the contribution is

$$\frac{(-1)^{\dim \mathfrak{a}_0^G}}{|W|} \int_{\widehat{T}^u} f^\vee(\chi)\, J_{M_0}(\chi^{-1}, \phi)\, d\chi.$$

- $M \supsetneq M_0$: $\tau = (M_0, \chi, r)$ with $r \in R_\chi^M$ elliptic in M, i.e. represented by $w \in W_\chi^M$ without non-zero fixed vector in $\mathfrak{a}_0^M$. Unramified twists by $i\mathfrak{a}_M^*$ act on such τ, so each orbit is a torus of dimension $\dim \mathfrak{a}_M^G$ in $\widehat{T}^u/W$. The contribution is an integral of $f^\vee(\chi_\tau) J_M(\tau^\vee, \phi)$ over these tori, with the constant of [Ar93].
- $M = G$: $\tau = (M_0, \chi, r)$ with $r \in R_\chi$ elliptic in G. These τ form a finite set, and contribute point masses $f^\vee(\chi_\tau) \operatorname{tr}\tau^\vee(\phi)$, with constants.

6 The main theorem

Theorem 6.1. *Let $\phi \in C_c^\infty(G_{\mathrm{ell}})$. With the normalizations of §11 assumed, for all $f \in \mathcal{H}(G,K)$,*

$$I(f,\phi) = c_G \int_{\widehat{T}^u} f^\vee(\chi)\, J_{M_0}(\chi^{-1},\phi)\, d\chi + \sum_{M \supsetneq M_0} \sum_{[\tau]} c_{M,\tau} \int_{i\mathfrak{a}_M^*/L_M^\vee} f^\vee(\chi_{\tau,\Lambda})\, J_M(\tau_\Lambda^\vee,\phi)\, d\Lambda,$$

where $[\tau]$ runs over the finitely many twist-orbits of $\tau = (M_0,\chi,r)$ with r elliptic in M, and the $c_{M,\tau}$ are constants from [Ar93]. Theorems A, B and C follow.

Proof outline. Equate Proposition 4.1 with the spectral side of §5.3.

- Theorem B: the normalized intertwining operators are holomorphic and unitary on the unitary axis, so $\chi \mapsto \mathcal{R}_{M_0}(\chi,P)$ is smooth on $\widehat{T}^u$. Since $I_P(\chi,\phi)$ has finite rank and depends smoothly on χ in the compact picture, $J_{M_0}(\chi^{-1},\phi)$ is smooth. Divide by ρ to get I_B.
- Theorem C: the same argument on each torus gives smooth densities. The list of §5.3 gives the supports.
- Theorem A: there are finitely many components, each with a smooth density on a compact torus, so the total variation is finite.

□

Remark 6.2. Theorem B shows that $I_B\rho$ is always smooth. In Route A, the T-independent part I has double poles on the root walls, cancelled by the double zeros of ρ [JC26, Rem. 6.2, (B.4)]. Route B explains this cancellation: it is the smoothness of a weighted character.

7 Comparison with Route A

Conjecture 7.1. *For regular unitary χ, the T-independent part $I(\chi,\phi)$ of the truncated kernel of [JC26, Prop. 4.2] equals $I_B(\chi,\phi)$. Equivalently,*

$$I(\chi,\phi)\,\rho(\chi) = c_G\, J_{M_0}(\chi^{-1},\phi).$$

Why it should hold. Both sides are linear in ϕ and determined by their pairings with all $f^\vee$, and Theorem 4.5(a) of [JC26] gives $I(f,\phi)$ as $\int f^\vee I^T\, d\mu$ for T large. If the oscillatory part of I^T pairs to zero in the limit, which happens for $PGL(2)$ and, numerically, for $PGL(3)$ ([JC26, App. B]), then $\int f^\vee I\, d\mu = I(f,\phi)$. Comparing with Theorem 6.1 for all f then gives Conjecture 7.1 when there is no singular part, because the $f^\vee$ are dense in $C(\widehat{T}^u/W)$. So the conjecture is already a consequence of the PGL(2) results of [JC26] and Theorem 6.1, and of the PGL(3) computation, once the normalizations are fixed. The real content is a direct proof that identifies the two constructions.

The two truncations. Arthur's $u(x,T)$ and the set $G(T)$ of Route A are both defined by the convex hull of $W \cdot T$. Route A's vertex terms $\psi(T_Q)R_Q$ [JC26, Lemma 3.5A] are periodic versions of Arthur's $e^{\Lambda(T_Q)}c_Q(\Lambda)/\theta_Q(\Lambda)$. In rank one, the constant term at $\Lambda = 0$ of $\sum_Q e^{\Lambda(T_Q)}c_Q(\Lambda)\theta_Q(\Lambda)^{-1}$ is a logarithmic derivative of an intertwining operator. This is the mechanism that produces a weighted character, and it is visible in [JC26, (6.5)].

What (R) means here. Hypothesis (R) of [JC26] asks that each oscillatory term of I^T, times ρ, be smooth. Theorem C says that only elliptic data on non-trivial R-groups can leave a singular part. For $PGL(3)$ all R-groups are trivial, and the poles found in [JC26, App. B] on the subtori $z_i^3 = 1$ must cancel in the limit, as observed. The weaker condition (R′) of [JC26, Rem. 2026.20] is a candidate for the analytic shadow of "no elliptic data".

8 $SL(n)$ and endoscopy

For $G = SL(n)$ and κ an unramified character of $PGL(n, F)$ of order $m \mid n$, the κ-orbital integrals and the elliptic endoscopic groups H_m are as in [JC26, §5]. Arthur's local character relations [Ar96] and his transfer of weighted orbital integrals [Ar99] relate the elliptic virtual characters $\tau = (M_0, \chi, r)$ of §5.3 to the stable characters of the H_m: the elliptic element $r \in R_\chi$ corresponds to a κ of order equal to that of r. Combined with Theorem 6.1, this would prove Theorem D and remove the hypothesis of [JC26, Prop. 5.2]. The stabilization of the local trace formula (Arthur's stable trace formula papers; see also Peng's explicit stable local trace formula [Pe16]) is the natural framework.

Status 8.1. This section is an outline. For κ of order n the endoscopic group is a torus and the result is [JC26, Prop. 5.1].

9 Examples

9.1 $PGL(2)$

Here $Z(G)$ is connected, all R-groups are trivial, and Theorem 6.1 has only the M_0 term. Theorem B says that $I_B\rho$ is the smooth function $-\frac{1}{2}J_A(\chi^{-1}, \phi)$. By [JC26, Thm. 6.1, Rem. 6.2], for γ of type (U_r),

$$I(\chi, \gamma)\rho(\chi) = \frac{Q}{2} \cdot \frac{|B_r|(2 - 2\cos\theta) + q^r(2\cos\theta - 2q^{-1})}{1 - 2q^{-1}\cos\theta + q^{-2}} \qquad (y = e^{i\theta}),$$

which is smooth, as Theorem B predicts. Conjecture 7.1 identifies this with a weighted character.

Status 9.1. Next step: compute $J_A(\chi^{-1}, \phi)$ directly, for ϕ the normalized characteristic function of γK_1. This is a trace on the $(q+1)$-dimensional space of level-one vectors, which can be done numerically, and the result compared with the display above. This will fix the constant c_G and the sign.

9.2 $SL(2)$

At the unramified quadratic character χ_0, $R_{\chi_0} = \{1, s\}$ and s acts by -1 on $\mathfrak{a}_0$, so s is elliptic. The corresponding virtual character is $\tau = \pi^+ - \pi^-$, where $I(\chi_0) = \pi^+ \oplus \pi^-$ and π^+ contains the spherical vector. So $\operatorname{tr}\tau(f) = f^\vee(\chi_0)$, and the $M = G$ term of Theorem 6.1 is a point mass at χ_0 with weight $c_\tau \operatorname{tr}\tau^\vee(\phi)$. For ϕ concentrated near an unramified elliptic γ of depth 0, $\operatorname{tr}\tau^\vee$ is the κ-character, and [JC26, Prop. 6.4] gives the weight $\frac{1}{2}$. This fixes c_τ for $SL(2)$, and the result agrees with Route A.

9.3 $PGL(3)$

All R-groups are trivial, so Theorem C predicts no singular part, in agreement with the numerical result of [JC26, App. B]. Theorem B predicts that $I\rho$ is smooth, which agrees with the closed form [JC26, (B.4)]:

$$I(\chi,\gamma)\rho(\chi) = \frac{(q-1)^2(q^2+q+1)}{2q^6}\cdot\frac{2(q^2+q+1)-q\sum_{\alpha\in\Phi}z^\alpha}{\prod_{\alpha\in\Phi}(1-q^{-1}z^\alpha)}.$$

Conjecture 7.1 predicts that this is $c_G J_{M_0}(\chi^{-1},\phi)$ for ϕ concentrated near γ.

9.4 $SL(3)$: a prediction

Let $G = SL(3)$, and let γ be regular elliptic in the norm-one torus of the unramified cubic extension. Up to W, the unitary unramified characters with non-trivial R-group and an elliptic element in it are the points

$$\chi_\zeta = (1,\zeta,\zeta^2) \qquad (\zeta^3=1,\ \zeta\neq 1),$$

in coordinates where χ is a triple modulo scalars (as for the dual group $PGL(3,\mathbb{C})$). At χ_ζ a 3-cycle in W fixes χ and acts on $\mathfrak{a}_0$ without fixed vectors. Here $R_{\chi_\zeta}\cong\mathbb{Z}/3$, $I(\chi_\zeta)$ has three constituents, and there are two elliptic virtual characters, attached to the two non-trivial elements of R. The Satake parameter of χ_ζ is $\mathrm{diag}(1,\zeta,\zeta^2)$, which is also the parameter χ_κ attached to the characters κ of order 3 in [JC26, Prop. 5.1].

Prediction. The spectral measure of O_γ on $SL(3)$ is $I\,d\mu$ plus a point mass at χ_ζ. Its weight is the sum of the two κ-contributions, $\frac{1}{3}(\Delta_\kappa^{-1}+\Delta_{\kappa^2}^{-1})c_S$ in the notation of [JC26, (5.4A), Prop. 5.1], which is $\frac{2}{3}$ if all transfer factors and c_S equal 1. There are no singular contributions on one-dimensional subtori: those would require an elliptic element of an R-group for $M\cong GL(2)$, and the R-groups of unramified principal series of $GL(2)$ are trivial.

Status 9.2. To be tested numerically on the building of $SL(3)$, by adapting the $PGL(3)$ code of [JC26, App. B] (`examples/pgl3/`) to vertices of one type and to the cubic norm-one torus. The predicted mass $\frac{2}{3}$ depends on normalizations that have not been checked.

10 Literature review

The first version of the companion paper was published in 1992 [Jo92] and corrected in 1997. Arthur's local trace formula and his theory of elliptic tempered characters appeared in between. So we divide the literature at 1997. §10.1 describes the tools available when the earlier versions were written, and §10.2 describes what came later and bears on Route B.

10.1 Before 1997

Spherical harmonic analysis. Satake identified the spherical Hecke algebra with W-invariant functions on the torus of unramified characters [Sa63]. Macdonald gave the explicit formulas for the spherical function and the spherical Plancherel measure [Ma71]. Casselman re-derived Macdonald's formula from intertwining operators and the asymptotics of matrix coefficients [Cas80], using the theory of Jacquet modules in his notes [Cas95]. Cartier's survey [Ca79] collects the unramified theory. Bernstein and Zelevinsky's geometric lemma [BZ77] describes Jacquet modules of induced representations.

Intertwining operators and R-groups. Knapp and Stein introduced the R-group governing the reducibility of induced representations for real groups [KS71], and Silberger proved the p-adic analogue of their dimension theorem [Si78]. Keys computed the R-groups of the unramified principal series of Chevalley groups [Ke82]. Langlands' normalizing factors for intertwining operators, built from L-functions, go back to his work on Eisenstein series [La76]. Van Dijk's formula for induced characters [vD72] gives the vanishing of characters of cuspidal functions on properly induced representations.

Arthur's trace formula and (G, M)-families. The invariant trace formula [Ar81, Ar88] introduced (G, M)-families, weighted orbital integrals and weighted characters, and the splitting formulas. Arthur studied weighted characters as functions of the representation in [Ar89b]. The local trace formula was announced in [Ar89a] and proved in [Ar91]. Arthur then reorganized its spectral side in terms of elliptic tempered virtual characters [Ar93], and studied the Fourier transforms of weighted orbital integrals [Ar94]. His paper on local character relations [Ar96] relates these objects to endoscopic transfer.

Elliptic harmonic analysis and endoscopy. Kazhdan's cuspidal geometry [Ka86] and the trace Paley–Wiener theorem of Bernstein, Deligne and Kazhdan [BDK86] control the elliptic part of harmonic analysis. Labesse and Langlands worked out L-indistinguishability and κ-orbital integrals for $SL(2)$ [LL79], and Langlands and Shelstad defined transfer factors [LS87]. Waldspurger computed the spherical orbital integrals of $GL(N)$ spectrally [Wa89a], by the truncation method that Route A follows.

10.2 After 1997

Normalization and transfer of weighted characters. Arthur proposed a canonical normalization of weighted characters, and a conjectural transfer identity for them [Ar98]. He then studied the transfer of weighted orbital integrals [Ar99], and stabilized the trace formula in a series of three papers [ArSTF]. The stabilization was completed for twisted groups by Mœglin and Waldspurger [MW16]. Arthur's classification of representations of classical groups [Ar13] uses the stabilized local and global trace formulas.

The fundamental lemma. Ngô's proof of the fundamental lemma [Ng10], combined with earlier reductions (see [JC26, §8]), made the endoscopic statements of §8 unconditional at the level of orbital integrals.

Elliptic representations. Reeder related the Euler–Poincaré pairing of Iwahori-spherical representations to elliptic representations of Weyl groups [Re01]. Opdam and Solleveld computed Ext groups of tempered representations of affine Hecke algebras, recovering Arthur's elliptic pairing [OS13]. These results describe exactly the elliptic virtual characters with Iwahori-fixed vectors, a class containing those that appear in Theorem C.

The Plancherel formula. Waldspurger's exposition of Harish-Chandra's Plancherel formula [Wa03] supplies complete proofs of the harmonic analysis behind Arthur's spectral side.

Variants of the local trace formula. Waldspurger proved a local trace formula adapted to the Gross–Prasad conjecture [Wa10]. Beuzart-Plessis proved one for the unitary Gan–Gross–Prasad conjecture [BP20]. Peng made the spectral side of the stable local trace formula explicit when one test function is cuspidal [Pe16]. H. Li proved a local trace formula for infinitesimal symmetric spaces [Li21].

Route A. The companion paper [JC26] revises [Jo92] by the truncation method, with the regularity hypothesis (R), and with examples in ranks one and two.

Status 10.1. The entries cited in this section have not yet all been checked against publisher records. The page ranges of [ArSTF], [MW16] and [Wa10] in particular are to be verified.

11 Status and what remains

(N1) *Normalizing factors.* Check that (3) satisfies Arthur's conditions for normalizing factors and agrees with [Ar98] in the unramified case (non-split groups included).

(N2) *Arthur's formula.* Fix the precise statements, constants and measures of the geometric and spectral expansions and of the elliptic form from [Ar91, Ar93], and the splitting formula (7) in the local setting.

(N3) *Lemma 5.3.* Supply a reference or a complete proof.

(N4) *Theorem 6.1.* Write the proof in full, including the smoothness of $\chi \mapsto J_{M_0}(\chi, \phi)$ in the compact picture.

(N5) *Rank one.* Compute $J_A(\chi, \phi)$ for $PGL(2)$ and $SL(2)$ numerically, and compare with [JC26, Thm. 6.1, Prop. 6.4]. This fixes c_G and c_τ.

(N6) *Conjecture 7.1.* Give a direct proof, at least in rank one.

(N7) $SL(3)$. Run the numerical test of the prediction.

(N8) *Theorem D.* Write §8 in full.

A Glossary

This glossary gives short definitions of the terms used in this paper, with pointers to where they are used. Terms already in the glossary of [JC26] are repeated only briefly.

Unramified group; hyperspecial K
A reductive group over F, quasi-split and split over an unramified extension. K is the stabilizer of a hyperspecial vertex of the building; for split G, $K = G(\mathcal{O})$.

Spherical function Γ_χ; Satake transform $f^\vee$
Γ_χ is the bi-K-invariant matrix coefficient of the spherical vectors of $I_B(\chi)$. The Satake transform is $f^\vee(\chi) = \operatorname{tr} I_B(\chi, f)$ for $f \in \mathcal{H}(G, K)$, a W-invariant Laurent polynomial in χ.

Plancherel measure $d\mu = \rho\, d\chi$
The measure on $\widehat{T}^u/W$ for which $f \mapsto f^\vee$ is an isometry from $L^2(K\backslash G/K)$ (Macdonald). ρ is proportional to $|c(\chi)|^{-2}$.

Principal series $I_P(\chi)$; compact picture
Normalized induction from a character of $M_0(F)$, realized on functions on K so that all $I_P(\chi)$ share one space.

Intertwining operator $J_{P'|P}(\chi)$
The operator $I_P(\chi) \to I_{P'}(\chi)$ given by integration over $N_{P'} \cap N_{\bar{P}}$, meromorphic in χ (§2.2).

Normalizing factor; normalized operator $R_{P'|P}$
A scalar $r_{P'|P}(\chi)$ with $R = r^{-1}J$ unitary and holomorphic on the unitary axis and multiplicative. In the unramified case it is the product of $L(0, \chi_\alpha)/L(1, \chi_\alpha)$ over the relevant roots, and R fixes the spherical vector (4).

(G, M)-family
A family $\{c_Q(\Lambda)\}_{Q \in \mathcal{P}(M)}$ of smooth functions that agree on common walls. Its limit $c_M = \lim_{\Lambda \to 0} \sum_Q c_Q(\Lambda)\theta_Q(\Lambda)^{-1}$ exists. For a constant family $c_M = 0$ unless $M = G$ [Ar81].

Weighted orbital integral $J_M(\gamma, f)$
An orbital integral against the weight function v_M attached to a (G, M)-family. It is not invariant unless $M = G$. These are the terms of the geometric side of Arthur's trace formulas.

Weighted character $J_M(\pi, f)$
$\operatorname{tr}(\mathcal{R}_M(\pi, P)I_P(\pi, f))$, with $\mathcal{R}_M$ the limit of the (G, M)-family of normalized intertwining operators (§2.3). These are the terms of the spectral side. They depend on the normalization.

Splitting formula
The expression (7) of a weighted character of $f_1 \otimes f_2$ through weighted characters of f_1 and of f_2 on Levi subgroups, with coefficients $d_M^G(L_1, L_2)$.

Local trace formula
Arthur's identity between the geometric expansion (5) and the spectral expansion (6) of the constant term $J(f)$ of a truncated integral of the kernel of $R(f)$ on $L^2(G(F))$, for $f \in C_c^\infty(G(F) \times G(F))$ [Ar91].

Truncation $u(x, T)$
The characteristic function of the compact set of $x \in G(F)$ whose Cartan component lies in the convex hull of $W \cdot T$. It is the analogue of the set $G(T)$ of Route A.

Regular elliptic element; G_{ell}
A regular semisimple element whose centralizer is an anisotropic torus (modulo the center).

Cuspidal function
$\phi \in C_c^\infty(G(F))$ whose orbital integrals vanish on non-elliptic regular elements; equivalently, $\operatorname{tr} \sigma(\phi) = 0$ for properly induced σ. Functions supported on G_{ell} are cuspidal (Lemma 5.2).

R-group R_χ
W_χ/W_χ°, the stabilizer of χ in W modulo the subgroup generated by the reflections in roots α with $\chi_\alpha = 1$. It controls the reducibility of $I_P(\chi)$ (§2.4).

Elliptic element of an R-group
An $r \in R_\chi$ represented by w with no non-zero fixed vector in $\mathfrak{a}_0^G$ (or in $\mathfrak{a}_0^M$, for M).

$T_{\text{disc}}(M)$, $T_{\text{ell}}(M)$
Arthur's sets of virtual tempered characters: triples (L, σ, r) with σ square-integrable on a

Levi $L \subseteq M$ and r in the R-group of σ in M. $T_{\mathrm{ell}}(M)$ is the subset with r elliptic [Ar93].

Absolutely continuous and singular parts
The decomposition $dI_\phi = I_B\, d\mu + dI_\phi^{\mathrm{sing}}$ of the spectral measure (Theorems B and C).

κ-orbital integral; endoscopic group; transfer factor
See [JC26, §5]. For $SL(n)$, κ is a character of $PGL(n,F)$ trivial on the image of $SL(n,F)$. The elliptic endoscopic groups are the H_m, and transfer factors Δ relate their stable orbital integrals to κ-orbital integrals.

Fundamental lemma
The identity, for the spherical Hecke algebra, between κ-orbital integrals and stable orbital integrals on endoscopic groups. It is a theorem (Ngô [Ng10] and others).

Route A, Route B
Route A is the truncation method of [JC26]. Route B, this paper, specializes Arthur's local trace formula.

Hypothesis (R)
The regularity hypothesis of [JC26, §4.3]: each oscillatory coefficient of the truncated kernel, times ρ, is smooth. It is sufficient for the spectral identity but not necessary (§7).

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